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(APPROVED BY AICTE, NEW DELHI AND AFFILIATED TO ANNA UNIVERSITY, CHENNAI)

(ACCREDITED BY NAAC, ISO 9001:2015 CERTIFIED INSTITUTION)

RESEARCH BULLETIN

2022 - 2023

(MARCH 2022 - DEC 2023)



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



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Vision

To impart international quality education, promote collaborative research and graduate industry-ready engineers in the domain of Artificial Intelligence and Data Science to serve the society.

Mission

- Excel in Teaching-Learning process and collaborative Research by the use of modern infrastructure and innovative components.
- Establish an Artificial Intelligence and Data Science based centre of excellence to prepare professional technocrats for solving interdisciplinary industry problems in various applications
- Motivate students to emerge as entrepreneurs with leadership qualities in a societal centric programme to fulfil Industry and community needs with ethical standards.

Program Educational Objectives (PEOs)

After successful completion of the degree, the students will be able to

- **PEO 1.** Apply Artificial Intelligence and Data Science techniques with industrial standards and pioneering research to solve social and environment-related problems for making a sustainable ecosystem.
- **PEO 2.** Excel with professional skills, fundamental knowledge, and advanced futuristic technologies to become Data Scientists, Data Analyst Managers, Data Science leaders AI Research Scientists, or Entrepreneurs.

Program Specific Outcomes (PSOs)

After successful completion of the degree, the students will be able to:

- **PSO 1:** To apply analytic technologies to achieve at meaningful insight and observation from data to solve engineering problems
- **PSO 2:** To create and apply Artificial Intelligence and Data Science techniques to forecast future trends in the domain of Healthcare, Education, Agriculture, Manufacturing, Automation, Robotics, and Transport, etc
- **PSO 3:** To enrich the critical thinking skills in emerging technologies such as Hybrid Mobile application development, cloud technology stack, and cyber-physical systems with mathematical aid to foresee the research findings and provide the solutions

Advisory Board

1. **Dr.Prabhat Kumar**
Professor & Head, Department of Computer science
NIT Patna
2. **Dr. ManjuKhari,**
Associate Professor , School of Computer and System Sciences,
Jawaharlal Nehru University, New Delhi
3. **Dr. JanmenjoyNayak,**
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Maharaja Sriram Chandra BhanjaDeo(MSCB) University
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4. **Dr. Asit K Das,**
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IEST Shibpur, Howrah, India
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University of Fortaleza (UNIFOR), Brazil
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University of Essex, UK
10. **Dr. Ruben Gonzalez Crespo**
Professor of Computer Science and Artificial Intelligence,
International University of La Rioja
Spain
11. **Dr Noor Zaman Jhanji**
Director Center for Smart Society 5.0 [CSS5]
Associate Professor, Dept of CS,
Taylor's University
Malaysia

International Collaboration

1.Publications

Institute/University	Country	No. of Publication
Noroff university college, Kristiansand Shaqra University	Norway	7
University of Teramo, Teramo,	Italy	2
Asia University, China Medical University, Foshan University, Foshan, Nanjing University of Information Science and Technology, Nanjing, China	China	6
Chulalongkorn University, Bangkok	Thailand	1
King Saud University, Taif University, University of Sfax, Sfax, Tunisia, University of Jeddah, Qassim University	Saudi Arabia	8
Papua New Guinea University of Technology	Papua New Guinea	1
Taylor's University, Xiamen University	Malaysia	5
Sejong University, Soonchunhyang University, Chung-Ang University, Sun Moon University, Zhuji affiliated hospital of Shaoxing University, Hanshin University, Hanshin University, Kyungpook National University, Chung-Ang University, Chung- Ang University	South Korea	17
Lebanese French University, Erbil,	Iraq	6
International University of La Rioja	Spain	3
RMIT University	Australia	2
University of Central Punjab, Lahore	Pakistan	1
Hamad Bin Khalifa University, Doha,	Qatar	1
Cairo University, University of Tartu	Egypt	2
Science and Research Branch, Islamic Azad University, Tehran,	Iran	1
Hamad Bin Khalifa University, Doha	Qatar	1
Halic University	Turkey	1
University of Maryland, Baltimore County	USA	1

2. MOU

1. Ubiquitous Computing and Security (UCS) Lab
Seoul National University of Science and Technology, Korea

2. MOBILE GRID AND CLOUD COMPUTING LAB, Department of Information Communication Engineering, Hankuk University of Foreign Studies, Korea

Scope

- Collaborative publication, Joint projects, organizing bilateral programmes



UCS lab



UCS lab team

3. Taylor's University, Malaysia

Dr Noor ZamanJhanjhi Associate Professor, School of Computer Science and Engineering, Faculty of Innovation and Technology, | Director Center for Smart society 5.0 and Cluster Head for Cyber security cluster, Taylor's University, Malaysia.

Scope

- Collaborative publication, joint projects, organizing bilateral programmes



3. Ph.D Holders & Honorary professor



Dr.M.Kaliappan
Professor and Head

Area of Interest: Artificial Intelligence ,
Data Analytics

Journal Publication: SCIE : 25, SCOPUS:03

Conference Publications: 15

Book Chapter: 05

Patent: 04

H-Index:17

Pursuing Post Doctoral - Federal University of
Ceara, Brazil

IEI sponsored project completed : 01

Theme: Deep learning



Dr.S.Vimal
Associate Professor

Area of Interest: Artificial Intelligence ,
Machine learning, Data Analytics

Journal Publication: SCIE : 107, SCOPUS:10

Conference Publications: 30

Book Chapter: 10

Books Published :03

Patent: 07

H-Index:25

Pursuing Post Doctoral - Federal University of
Ceara, Brazil



HONORARY PROFESSOR

Dr.Tu Nguyen,

Director of Intelligence System lab,
Department of Computer Science,
Kennesaw State University, Georgia, USA

Scope

- **Promote collaboration in research and academic activities**

Anna University, Chennai Recognised Supervisors

The Department has applied for Research Centre Recognition under Anna University in Dec 2022(Ref.No:16IR2331)

SI.NO	Name	Supervisor Number
1.	Dr.M.Kaliappan	2740025
2	Dr.S.Vimal	3840016

4. JOURNAL PUBLICATIONS

1.	SohailSaif, PriyaDas, SuparnaBiswas, ManjuKhari, VimalShanmuganathan ," HIIDS: Hybrid intelligent intrusion detection system empowered with machine learning and metaheuristic algorithms for application in IoT based healthcare" Microprocessors and Microsystems, Available online 12 August 2022, 104622, Elsevier,IF 3.503, Accepted for publication
2.	Priyanka chugh Meenu Gupta, S. Indu, Gopal Chaudhary, Manju Khari*, Vimal Shanmuganathan ," ADVANCED ENERGY EFFICIENT PEGASIS BASED ROUTING PROTOCOL FOR IoT APPLICATIONS," Microprocessors and Microsystems ,Available online 21 November 2022, 104727, Elsevier,IF 3.503.
3.	Azhagu Jaisudhan Pazhani.A, P. Gunasekaran, Vimal Shanmuganathan , Sangsoon Lim, Kaliappan Madasamy, Rajesh M, Amit Verma,"Peer-Peer communication using Novel Slice Handover Algorithm for 5G Wireless Networks", Journal of Sensor and Actuator Networks, MDPI, ESCI Indexed, Dec 2022
4.	P. Suresh Kumar, Pandit Byomakesha Dash, B. Kameswara Rao, S. Vimal & Khan Muhammad ,"Early Detection of Chronic Obstructive Pulmonary Disease Using LSTM-Firefly Based Deep Learning Model", pp 247-267, DOI https://doi.org/10.1007/978-3-031-17544-2_11 , Nature-Inspired Optimization Methodologies in Biomedical and Healthcare. Intelligent Systems Reference Library, vol 233. Springer, Cham. https://doi.org/10.1007/978-3-031-17544-2_11 , Nov 2022, Scopus Indexed.
5.	Janmenjoy Nayak , H. Swapnarekha , Bighnaraj Naik , Vimal Shanmuganathan ,Gaurav Dhiman ," 25 YEARS OF PARTICLE SWARM OPTIMIZATION: FLOURISHING VOYAGE OF TWO DECADES", Archives of Computational Methods in Engineering, Springer, , Oct 2022,SCIE Indexed, IF 8.171-ACCEPTED
6.	Antony Taurshia , Jasper W Kathrine , Alireza Sourì , S E Vinodh, S Vimal , Kuan-Ching Li, S Sudhakar Ilango "Software-Defined Network aided Lightweight Group Key Management for resource-constrained Internet of Things devices" , Sustainable Computing: Informatics and Systems, Elsevier, Oct 2022,SCIE Indexed, IF 4.923-ACCEPTED & online

7.	Debasmita Mishra , Bighnaraj Naik, Janmenjoy Nayak , Alireza Sourì , Pandit Byomakesha Dashe, S.Vimal , "Light gradient boosting machine with optimized hyperparameters for identification of malicious access in IoT network", Digital Communications and Networks, Elsevier, Oct 2022, SCIE Indexed, IF 6.348.
8.	Rebin M. Ahmed,polla Fattah, Abeer Alsadoon, Nebojsa Bacanin, Seyedali Mirjalili, S. Vimal , Amit Chhabra, "Kurdish Handwritten Character Recognition using Deep Learning Techniques", Gene Expression Patterns, Elsevier, Sep 2022, SCIE Indexed, IF1.448.
9.	L.Ganesan, C.Umarani, M.Kaliappan, S.Vimal , Seifedine Kadry, Yunyoung Nam, "Texture Image Analysis for Larger Lattice Structure using Orthogonal Polynomial framework", Information Technology and Control, Journal of Kaunas University of Technology,Sep 2022, IF:1.228
10.	Muhammad Bilal,Raja Majid Mehmood; S.Vimal ; Seong-Whan Lee,"EEG-based Affective State Recognition from Human Brain Signals by Using Hjorth-Activity", Measurement, Elsevier, Aug 2022, IF 5.131.
11.	Xu, Xiaolong; Liu, Zhongjian ; Bilal, Muhammad; Vimal, S ; Song, Houbing , "Computation Offloading and Service Caching for Intelligent Transportation Systems with Digital Twin,", IEEE Intelligent Transportation Systems Transactions, IEEE, SCIE Indexed, July 2022,Impact Factor-9.5
12.	Yongzhao Xu , Paulo C. S. Barbosa , Joel S. Da Cunha Neto , Lijuan Zhang , S.Vimal , Victor Hugo C. De Albuquerque, Subbulakshmi p. , Development of intelligent and integrated technology for pattern recognition in EMG signals for robotic prosthesis command, Expert Systems , Wiley Publication, SCIE Indexed, July 2022
13.	Maryam T. Abdulkhaleq, Sazan L. Ali Tarik A. Rashid, Abeer Alsadoon, Mokhtar Mohammadi, Jaza M. Abdullah, Amit Chhabra, Rawshan N. Othman, Hadil A. Hasan, Sara Azad, Naz A. Mahmood, Sivan S. Abdalrahman, S. Vimal ,Hezha O. Rasul, Nebojsa Bacanin,"Harmony Search: Current Studies and Uses on Healthcare Systems", Artificial Intelligence In Medicine,Elsevier,June 2022, SCIE Indexed, IF: 7.011
14.	Farhana Ajaz and Mohd Naseem, Sparsh Sharma, Gaurav Dhiman, Mohammad Shabaz, S. Vimal ,"Architecture and routing protocols for internet of vehicles: a review" Int. J. Ad Hoc and Ubiquitous Computing, Vol. 40, Nos. 1/2/3, 2022
15.	Barzan Hussein Tahir ,Tarik A. Rashid , Hafiz Tayyab Rau, Nebojsa Bacanin,Amit Chhabra, S. Vimal ,Zaher Mundher Yaseen," Improved Fitness Dependent Optimizer for Solving Economic Load Dispatch Problem" Computational Intelligence and Neuroscience,Impact Factor-3.6, SCIE Indexed, June 2022
16.	Akhtar Badshah, Muhammad Waqas, Ghulam Abbas, Fazal Muhammad, Ziaul Haq Abbas, Dr. Muhammad Bilal S Vimal , "LAKE-BSG: Lightweight Authenticated Key Exchange Scheme for Blockchain-enabled Smart Grids", Sustainable Energy Technologies and Assessments, Volume 52, Part C, August 2022, 102248, SCIE Indexed, Impact Factor 5.3, May 2022
17.	A.AlfredRajaMelvin, G.JaspherWKathrine, Subbulakshmi Pasupathi, S. Vimal , R.Naganathan An AI Powered System Call Analysis with Bag of Word Approaches for the detection of Intrusions and Malware in ADFA and VMM Malware Attack Dataset" Expert Systems , Wiley Publication, SCIE Indexed , May 2022. A. Alfred Raja Melvin, G. Jaspher W Kathrine, Dr. Subbulakshmi Pasupathi, S. Vimal, and Dr. R. Naganathan (2021), An AI Powered System Call Analysis with Bag of Word Approaches for the detection of Intrusions and Malware in ADFA and VMM Malware Attack Dataset, Emerging Trends in Artificial Intelligence for Intelligent Industrial Applications, 2021;00:1-6.
18.	M. Kaliappan G. Karpagarajesh,R. Santhana Krishnan , Y. Harold Robinson, S. Vimal , S. Thamizharasan, P. Subbulakshmi , "Comparative analysis of FSO, OFC and diffused channel links in Photonics using Artificial Intelligence based S-BAND,C-BAND and L-BAND of the attenuation metrics,"Sustainable Energy Technologies and Assessments, Elsevier, IF: 5.35, May 2022
19.	Xu, Xiaolong, Liu, Zhongjian ; Bilal, Muhammad; Vimal, S ; Song, Houbing, " Task Offloading and Caching with Decision Theory for Intelligent Transportation with Digital Twin",IEEE Transactions on Intelligent Transportation Systems, IF 6.492, April 2022
20.	Book: BIOINFORMATICS AND MEDICAL APPLICATIONS: Big Data Using Deep Learning Algorithms Edited by A. Suresh, S. Vimal , Y. Harold Robinson, Dhinesh Kumar Ramaswami and R. Udendhran Copyright: 2022 Expected Pub Date:2022/05/30 ISBN: 9781119791836 Hardcover 315 pages

	ISBN-13: 978-1119791836 ISBN-10: 1119791839 Publisher: Wiley
21.	Dilip Sisodia Manoj Diwakar, Prabhishek Singh , Achyut Shankar,Soumya Ranjan Nayak, Janmenjoy Nayak, S.Vimal ,Ravinder Singh,"Directive Clustering Contrast based Multi-Modality Medical Image Fusion for Smart Healthcare System", Network Modeling Analysis in Health Informatics and Bioinformatics, Springer, March 2021, Scopus
22.	Ghazaala Yasmin, ASIT KUMAR DAS, Janmenjoy Nayak, S Vimal , Soumi Dutta," A rough set theory and deep learning based predictive system for gender recognition using audio speech, Soft Computing, April 2022, IF:3.643
23.	Dilip Sisodia Manoj Diwakar,Prabhishek Singh, Achyut Shankar,Soumya Ranjan Nayak,Janmenjoy Nayak, S. Vimal , Ravinder Singh," Directive clustering contrast-based multi-modality medical image fusion for smart healthcare system" Network Modeling Analysis in Health Informatics and Bioinformatics, Springer, March 2022, Scopus Indexed
24.	Janmenjoy Nayak, Saroj K. Meher, Alireza Souri, Bighnaraj Naik, S. Vimal , "Extreme Learning Machine and Bayesian Optimization-driven Intelligent Framework for IoMT Cyber-attack detection", Journal of Supercomputing, Springer April 2022, SCIE indexed ,IF 2.6
25.	Mohit Kumar, Priya Mukherjee ,Sahil Verma ,Kavita ,NZ Jhanjhi , S Vimal , Bandar Almutain ,"A Novel SDN-Based Security Framework for Wireless Sensor Networks Using TDCNN And PGF-ECC", Human-centric Computing and Information Sciences, March 2022, SCIE indexed ,IF 5.9 - Accepted
26.	Ankit Agrawal, Rajiv Singh, Manju Khari,- S.Vimal , Sangsoon Lim, "Autoencoder for Design of Mitigation Model for DDOS Attacks viaM-DBNN" Wireless Communications and Mobile Computing, SCIE indexed, March 2022,IF 2.336
27.	Vatsala Anand, Sheifali Gupta, Deepika Koundal, Soumya Ranjan Nayak, Janmenjoy Nayak, S. Vimal Multi-Class Skin Disease Classification using Transfer Learning Model, International Journal on Artificial Intelligence Tool, March 2022,SCIE Indexed
28.	Hanzhi Yan, Muhammad Bilal, Xiaolong Xu, S Vimal , "Edge Server Deployment for Health Monitoring with Reinforcement Learning in Internet of Medical Things", IEEE Transactions on Computational Social Systems, March 2022,SCIE Indexed

5. CONFERENCE PUBLICATIONS

Saubhik Bandyopadhyay Pushan Kumar Dutta, Abdul Rehman Javed,Kassian T.T. Amesho, Abhimanyu Bhowmick, Rahul P, Madhushree Sannigrahi , **Vimal Shanmuganathan** ,"A new type of audio steganography with increased privacy using different ratios of LSB embedding", 6th IET Smart Cities Symposium, IET, 6-9 December 2022 ,Zain – e-Learning Center – University of Bahrain, Bahrain, Accepted for publication

Geetanjali Bhoi, Bighnaraj Naik, Etuari Oram, **S.Vimal**, Gravitational Search Optimized Light Gradient Boosting Machine for Identification of malicious access in IoT Network" International Conference on Computational Intelligence in Pattern Recognition (CIPR 2022) IEST Shibpur, India April 23-24, 2022, Springer

6. BOOK/BOOK CHAPTER PUBLICATIONS

Book Title: Computational Intelligence in Pattern Recognition
Book Subtitle Proceedings of CIPR 2022

Editors:Asit Kumar Das, Janmenjoy Nayak, Bighnaraj Naik, S. Vimal, Danilo Pelusi Series Title: Lecture Notes in Networks and Systems DOI: https://doi.org/10.1007/978-981-19-3089-8 Publisher: Springer Singapore eBook Packages: Intelligent Technologies and Robotics, Intelligent Technologies and Softcover ISBN:978-981-19-3088-1 eBook ISBN:978-981-19-3089-8 Series ISSN:2367-3370 Series E-ISSN:2367-3389 Edition Number:1 Number of Pages: XXVIII, 665 Number of Illustrations: 77 b/w illustrations, 265 illustrations in colour Topics : Computational Intelligence, Computer Imaging, Vision, Pattern Recognition and Graphics, Artificial Intelligence, Mobile and Network Security
Book Title: Computational Intelligence in Data Mining Book Subtitle: Proceedings of ICCIDM 2021 Editors Janmenjoy Nayak, H.S. Behera, Bighnaraj Naik, S. Vimal, Danilo Pelusi Series TitleSmart Innovation, Systems and Technologies DOI https://doi.org/10.1007/978-981-16-9447-9 Publisher: Springer Singapore eBook Packages Intelligent Technologies and Robotics, Intelligent Technologies and Robotics (R0) Hardcover ISBN978-981-16-9446-2 eBook ISBN978-981-16-9447-9 Series ISSN2190-3018
Series E-ISSN2190-3026 Edition Number1 Number of Pages:XXVI, 764 Number of Illustrations76 b/w illustrations, 269 illustrations in colour Book: BIOINFORMATICS AND MEDICAL APPLICATIONS: Big Data Using Deep Learning Algorithms Edited by A. Suresh, S. Vimal, Y. Harold Robinson, Dhinesh Kumar Ramaswami and R. Udendhran ISBN: 9781119791836 Hardcover 315 pages ISBN-13: 978-1119791836 ISBN-10: 1119791839 Publisher: Wiley



Microprocessors and Microsystems

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In Press, Journal Pre-proof [What's this?](#)

HIIDS: Hybrid intelligent intrusion detection system empowered with machine learning and metaheuristic algorithms for application in IoT based healthcare

Sohail Saif¹ , Priya Das¹ , Suparna Biswas¹ , Manju Khari²  , **Vimal Shanmuganathan³**  [Show more](#)  Share  Cite<https://doi.org/10.1016/j.micpro.2022.104622>[Get rights and content](#)

Abstract

This paper presents machine learning and metaheuristic algorithms based hybrid intelligent Intrusion Detection System (HIIDS) for Internet of Things based applications such as healthcare. In IoT based smart healthcare, biomedical sensors sense vital health parameters which are sent to the cloud server for storage and analysis. Health data saved as Electronic Health Record (EHR) is privacy and security sensitive. This work focuses on the detection of security attacks on cloud servers through anomaly based intrusion detection. Popular NSL-kDD dataset containing 41 features with 125,973 samples have been utilized for performance evaluation of proposed HIIDS. To reduce computation cost, metaheuristic algorithms such as Particle Swarm Optimization (PSO), Genetic Algorithm (GA), and Differential Evolution (DE) are used for best feature selection and supervised learning algorithms such as Known Nearest Neighbor (kNN), Decision Tree (DT) are used for accurate classification of normal and attack class based on selected features. Also a hybrid approach has been presented for feature selection and classification. After dataset pre-processing using python, MATLAB 2019b is used to implement six variants of proposed hybrid algorithms combining GA, PSO, DE with kNN, DT. Performance evaluation has been done based on accuracy, execution time, memory usage and CPU utilization. GA-DT variant gives highest accuracy of 99.88%, 86.40%, 95.39%, 96.90%, 100% of accuracy for DoS, U2R, R2L, Probe and Normal class with the help of 8-10 features compared to other variants such as GA-kNN, PSO-kNN, PSO-DT, DE-kNN, DE-DT. Also outperforms similar state-of-the-art works in terms of classification accuracy, simulation results are given in support. Finally an IoT based healthcare architecture is designed using best performing hybrid GA-DT variant based HIIDS to detect and prevent malicious traffic.

Introduction

Internet of things plays a pivotal role in transforming almost all applications in smart ones whereas machine learning adds intelligence to them for efficient data handling [1]. Several applications associated with daily



Microprocessors and Microsystems

Available online 21 November 2022, 104727

In Press, Journal Pre-proof [What's this?](#)

ADVANCED ENERGY EFFICIENT PEGASIS BASED ROUTING PROTOCOL FOR IoT APPLICATIONS

Priyanka chugh¹ , Meenu Gupta² , Dr.S. Indu³, Gopal Chaudhary⁴ , Manju Khari⁵ ,
Vimal Shanmuganathan⁶  

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Abstract

The Scientific betterments in Wireless Sensor Networks (WSNs) technology have paved the way for the advancement of lower sized, reduced-cost, and highly-powered sensor nodes. The sensor nodes are consistently expected to work with limited power batteries, and their energy is the most precious asset in a WSN, proper usage of the total energy to perpetuate the Network's lifetime is the primary constraint and objective of the majority of the research works. As energy conservation and prolonging of network-lifetime have become a critical issue, a novel protocol for WSN based on an Advanced Fixed Path and Mobile Base Station energy-efficient PEGASIS based routing protocol is proposed in this research work. The proposed method focuses on the base station's specified mobility with a multiple-chain model to attain small size chains and reduce the single leader node load that directly affect the Network's energy conservation and lifetime. The experimental results based on average energy, residual energy, consumed energy, dead nodes, and alive nodes reflect the proposed method's efficiency to fulfill the above-said objectives. The proposed method is also compared with three current routing protocols, i.e., Energy Efficient Routing Algorithm with Mobile Sink Support for WSNs (EERCCSM), Cluster-Chain Mobile Agent Routing (CCMAR), and Energy-efficient Cluster-based Dynamic Routing Algorithm (ECDRA). This novelty and advancement of the base station will open new horizons for getting it placed much closer to the cluster leader to prolong Network's lifetime to a great extent in the future.

Section snippets

INTRODUCTION

WSNs are adapted to sense distinct parameters of the environment by adopting wireless sensor nodes [1]. One of the noteworthy improvements in WSN-based smart healthcare centers is the infusion of IoTs (the Internet of Things). The universal Network and extraordinary intelligence of the IoTs with remote detecting

Article

Peer–Peer Communication Using Novel Slice Handover Algorithm for 5G Wireless Networks

Azhagu Jaisudhan Pazhani.A ¹, P. Gunasekaran ¹, Vimal Shanmuganathan ² , Sangsoon Lim ^{3,*} ,
Kaliappan Madasamy ², Rajesh Manoharan ⁴ and Amit Verma ⁵

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Citation: Pazhani.A, A.J.; Gunasekaran, P.; Shanmuganathan, V.; Lim, S.; Madasamy, K.; Manoharan, R.; Verma, A. Peer–Peer Communication Using Novel Slice Handover Algorithm for 5G Wireless Networks. *J. Sens. Actuator Netw.* **2022**, *11*, 82. <https://doi.org/10.3390/jsan11040082>

Academic Editors: Jordi Mongay Batalla, Dilbag Singh, Robertas Damaševičius and Vijay Kumar

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Abstract: The goal of 5G wireless networks is to address the growing need for network services among users. User equipment has progressed to the point where users now expect diverse services from the network. The latency, reliability, and bandwidth requirements of users can all be classified. To fulfil the different needs of users in an economical manner, while guaranteeing network resources are resourcefully assigned to consumers, 5G systems plan to leverage technologies like Software Defined Networks, Network Function Virtualization, and Network Slicing. For the purpose of ensuring continuous handover among network slices, while catering to the advent of varied 5G application scenarios, new mobility management techniques must be adopted in Sliced 5G networks. Users want to travel from one region of coverage to another region without any fading in their network connection. Different network slices can coexist in 5G networks, with every slice offering services customized to various QoS demands. As a result, when customers travel from one region of coverage to another, the call can be transferred to a slice that caters to similar or slightly different requirements. The goal of this study was to develop an intra- and inter-slice algorithm for determining handover decisions in sliced 5G networks and to assess performance by comparing intra- and inter-slice handovers. The proposed work shows that an inter-slice handover algorithm offers superior quality of service when compared to an intra-slice algorithm.

Keywords: 5G; latency; bandwidth; radio; slicing; QoS; NFV; machine

1. Introduction

Since the introduction of the first generation of wireless technology, the telecommunications industry has grown at an exponential rate. The industry's growth can be attributed to an increase in customer demand for network services. Users requesting network services have a wide range of needs, including streaming videos and audio, sending and receiving multimedia files, and making online transactions. These services necessitate network providers providing high levels of QoS, while resourcefully consuming network resources.

In this regard, 5G networks use different kinds of technologies and Network Slicing to accommodate the varying needs of users. Network slicing is the rational division of a physical network structure into multiple autonomous virtual networks, each of which is designed to meet the needs of specific users. The use of radio resources is improved through network slicing, which ensures that only the resources required are readily available. Network slicing ensures that the networks make optimum use of network resources. In order to deliver acceptable QoS, mobility management is a major difficulty in 5G networks where continuous connectivity, smooth at cell borders, is anticipated.



Nature-Inspired Optimization Methodologies in Biomedical and Healthcare pp 247–267

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Early Detection of Chronic Obstructive Pulmonary Disease Using LSTM-Firefly Based Deep Learning Model

[P. Suresh Kumar](#) , [Pandit Byomakesha Dash, B. Kameswara Rao, S. Vimal](#) & [Khan Muhammad](#)

Chapter | [First Online: 15 November 2022](#)

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Abstract

Identifying Chronic Obstructive Pulmonary Disease (COPD) is essential for reducing mortality and cost burden. However, the population suffers from an underdiagnosis of chronic obstructive pulmonary disease. This chapter aims to create COPD detection models and assess the relative effectiveness of several modeling paradigms to discover the optimal model for the task on the dataset of 563

[Home](#) > [Archives of Computational Methods in En...](#) > [Article](#)Review article | [Published: 03 December 2022](#)

25 Years of Particle Swarm Optimization: Flourishing Voyage of Two Decades

[Janmenjoy Nayak](#) , [H. Swapnarekha](#), [Bighnaraj Naik](#),
[Gaurav Dhiman](#) & [S. Vimal](#)[Archives of Computational Methods in Engineering](#) **30**,
1663–1725 (2023)**492** Accesses | **4** Citations | [Metrics](#)

Abstract

From the past few decades many nature inspired algorithms have been developed and gaining more popularity because of their effectiveness in solving problems of distinct application domains.

Undoubtedly, Particle swarm optimization (PSO) algorithm is the most successful optimization algorithm among the available nature inspired algorithms such as simulated annealing, genetic algorithm, differential evolution, firefly, cuckoo etc., because of its high efficiency and capability to adjust in different dynamic environments. This year marks its 25th anniversary of PSO, one of the base inspirations for many modern-day metaheuristics



Sustainable Computing: Informatics and Systems

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Software-defined network aided lightweight group key management for resource-constrained Internet of Things devices

Antony Taurshia^a , G. Jaspher Willsie Kathrine^a , Alireza Souri^b , S.E. Vinodh^a , S. Vimal^c ,
Kuan-Ching Li^d , S. Sudhakar Ilango^e 

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Abstract

Software-Defined Network (SDN) and Network Function Virtualization (NFV) are key enablers for provisioning dynamic, virtual security services on demand. This article, presents a novel Group Key Management scheme for Low-Resource Devices (GKM-LRD) with SDN aided trusted key management server as a central server to provide key management service to groups formed in Internet of Things (IoT) applications. In addition to the group key management scheme, a lightweight packet key-based communication system is proposed to make the data traffic reliable. Packet keys are short sized keys acquiring the lifetime of a packet. Even though the key is short sized, the security strength comes from its short lifetime. The proposed GKM-LRD is evaluated in terms of computation load, communication load, storage and scalability in comparison with the existing group key management schemes. The computation load, communication load and storage load is reduced by 80%, 17.25% and 20% respectively in comparison with the existing system assuring the scheme as lightweight and applicable for IoT devices. The proposed packet key-based communication scheme is compared with an existing Diffie-Hellman based packet key system. The results show the system is comparatively lightweight with computation load reduced by 74.8%. The security analysis of the proposed scheme proves that the scheme ensures forward secrecy, backward secrecy and resistance to man-in-the-middle and replay attacks.

Introduction

The ubiquitous IoT applications demands the services of fog and cloud service providers to perform cost effective tasks [1]. In healthcare application cloud and fog services are leveraged for application partitioning and scheduling tasks to minimize energy consumption and application enhancement [2], [3]. In Internet of Vehicular Things (IoVT), the fog, cloud service is leveraged to provide mobility-aware services based on containers [4]. The fog cloud services are further enhanced with the inclusion of technologies like SDN and Intelligent Reflecting Surface (IRS) [5]. SDN is an emerging platform that uses software to program the



Light gradient boosting machine with optimized hyperparameters for identification of malicious access in IoT network

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Abstract

In this paper, an advanced and optimized Light Gradient Boosting Machine (LGBM) technique is proposed to identify the intrusive activities in the Internet of Things (IoT) network. The followings are the major contributions: i) An optimized LGBM model has been developed for the identification of malicious IoT activities in the IoT network; ii) An efficient evolutionary optimization approach has been adopted for finding the optimal set of hyper-parameters of LGBM for the projected problem. Here, a Genetic Algorithm (GA) with k-way tournament selection and uniform crossover operation is used for efficient exploration of hyper-parameter search space; iii) Finally, the performance of the proposed model is evaluated using state-of-the-art ensemble learning and machine learning-based model to achieve overall generalized performance and efficiency. Simulation outcomes reveal that the proposed approach is superior to other considered methods and proves to be a robust approach to intrusion detection in an IoT environment.



Kurdish Handwritten character recognition using deep learning techniques

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Abstract

Handwriting recognition is regarded as a dynamic and inspiring topic in the exploration of pattern recognition and image processing. It has many applications including a blind reading aid, computerized reading, and processing for paper documents, making any handwritten document searchable and converting it into structural text form. High accuracy rates have been achieved by this technology when recognizing handwriting recognition systems for English, Chinese Arabic, Persian, and many other languages. However, there is not such a system for recognizing Kurdish handwriting. In this paper, an attempt is made to design and develop a model that can recognize handwritten characters for Kurdish alphabets using deep learning techniques. Kurdish (Sorani) contains 34 characters and mainly employs an Arabic/Persian based script with modified alphabets. In this work, a Deep Convolutional Neural Network



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Texture Image Analysis for Larger Lattice Structure using Orthogonal Polynomial framework

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DOI: <https://doi.org/10.5755/j01.itc.51.3.29322>

Keywords: Texture Representation, Polynomials, Statistical Tests, Local and Global descriptors, Standard Texture Images

Abstract

An Orthogonal Polynomial Framework using 3×3 mathematical model has been proposed and attempted for the texture analysis by L. Ganesan and P. Bhattacharyya during 1990. They proposed this framework which was unified to address both edge and texture detection. Subsequently, this work has been extended for different applications by them and by different authors over a period of time. Now the Orthogonal Polynomial Framework has been shown effective for larger grid size of (5×5) or (7×7) or higher, to analyze textured surfaces. The image region (5×5) under consideration is evaluated to be textured or untextured using a statistical approach. Once the image region is concluded to be textured, it is proposed to be described by a local descriptor, called pro5num, computed by a simple coding scheme on the individual



Measurement

Volume 202, October 2022, 111738

EEG-based affective state recognition from human brain signals by using Hjorth-activity

Raja Majid Mehmood^{a, b} , Muhammad Bilal^c , S. Vimal^d , Seong-Whan Lee^e  ^a Information and Communication Technology Department, School of Computing and Data Science, Xiamen University Malaysia, Sepang 43900, Malaysia^b Department of Brain and Cognitive Engineering, Korea University, Anam-dong, Seongbuk-gu, Seoul 02841, South Korea^c Department of Computer Engineering, Hankuk University of Foreign Studies, Yongin-si 17035, South Korea^d Data Analytics Lab, Department of AI & DS, Ramco Institute of Technology, Rajapalayam 626 117, Virudhunagar, Tamil Nadu, India^e Department of Artificial Intelligence, Korea University, Anam-dong, Seongbuk-gu, Seoul 02841, South Korea

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Abstract

EEG-based emotion recognition enables investigation of human brain activity, which is recognized as an important factor in brain-computer interface. In recent years, several methods have been studied to find optimal features from brain signals. The main limitation of existing studies is that either they consider very few emotion classes or they employ a large feature set. To overcome these issues, we propose a novel Hjorth-feature-based emotion recognition model. Unlike other methods, our proposed method explores a wider set of emotion classes in the arousal-valence domain. To reduce the dimension of the feature set, we employ Hjorth parameters (HPs) and analyze the parameters in the frequency domain. At the same time, our study was focused to maintain the accuracy of emotion recognition for four emotional classes. The average accuracy was approximately 69%, 76%, 85%, 59%, and 87% for DEAP, SEED-IV, DREAMER, SELEMO, and ASCERTAIN, respectively. Results show that the features from HP activity with random forest outperforms all the classic methods of EEG-based emotion recognition.

Computation Offloading and Service Caching for Intelligent Transportation Systems With Digital Twin

Publisher: IEEE Cite This PDF

Xiaolong Xu ; Zhongjian Liu ; Muhammad Bilal ; S. Vimal ; Houbing Song All Authors

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Abstract

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IV. Problem Transformation and Algorithm Design

V. Simulation Results

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Metadata

Abstract:Mobile edge computing (MEC) provides a novel computing paradigm to satisfy the increasing computation requirements of mobile applications. In MEC-enabled intelligent transportation systems (ITS), the latency-sensitive computing tasks are offloaded to RSUs for execution, reducing the transmission latency compared with the cloud solutions. However, the repetitive executions of the same tasks whose outputs are dependent on the inputs lead to the extra system latency, an alternative is to cache the required services on RSUs in advance. The service requirements of latency-sensitive computing tasks are satisfied by jointly considering computation offloading and service caching. Besides, the digital twin (DT) is utilized to construct the virtual world reflecting the physical world in real-time to efficiently make offloading strategies. In this paper, a computation offloading and service caching method using decision theory in ITS with DT, named CODT, is proposed. Specifically, the computation offloading and service caching in ITS is modeled first with DT. Then, a mixed-integer nonlinear programming (MINLP) problem is formulated to minimize the system latency. Afterward, the decision theory is used to analyze the utilities of offloading strategies in different states of RSUs and make the optimal strategy. Finally, extensive simulations based on the real-world datasets demonstrate that the proposed CODT outperforms other baselines.

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Expert Systems / Volume 40, Issue 5 / e13109

ORIGINAL ARTICLE

Development of intelligent and integrated technology for pattern recognition in EMG signals for robotic prosthesis command

Yongzhao Xu, Paulo C. S. Barbosa, Joel S. da Cunha Neto, Lijuan Zhang, Vimal Shanmuganathan✉, Victor Hugo C. de Albuquerque✉, Subbulakshmi Pasupathi✉

First published: 02 August 2022

<https://doi.org/10.1111/exsy.13109>

Abstract

Prostheses play an important role in the rehabilitation of people who have suffered some type of amputation. However, due to its high-cost and high complexity in performing movements of everyday tasks, users of these prostheses may encounter many difficulties. Therefore, this work proposes the development of a future artificial intelligence technology based on a low-cost functional prosthesis prototype (manufactured in a 3D printer). In the present work, we describe an intelligent system that uses an artificial neural network to recognize patterns in muscle biopotential signals in order to control a prosthesis prototype in real time. Such a system is divided into three parts: the first that performs a human–machine integration through a graphical user interface; the second that performs the signal acquisition; the third that performs the training and generalization steps of the artificial neural network. The developed interface runs on a web application that has a database hosted in the cloud and in it the system user can: Acquisition of electromyography signals; Training phase of the artificial neural network; Sends the matrix of weights of the trained network to the microcontroller; Activates in the microcontroller, the state of action of the commands from the identified gestures. To compose the results of the present work, a search was initially carried out for the ideal parameters of the artificial neural network through signals obtained from 20 volunteers. In this step, it was possible to identify the topology that best classifies the signals of each gesture, as well as the investigation of the number of neurons in the hidden layer that causes a low generalization power due to overfitting. At the end of the project, it was possible to validate the use of the system with 15 new volunteers, and it was observed that in most cases, the performance of the commands in the prosthesis



Artificial Intelligence in Medicine

Volume 131, September 2022, 102348

Harmony search: Current studies and uses on healthcare systems

Maryam T. Abdulkhaleq^{a, k}, Tarik A. Rashid^a, Abeer Alsadoon^{b, c}, Bryar A. Hassan^d  , Mokhtar Mohammadi^e, Jaza M. Abdullah^{f, g}, Amit Chhabra^h, Sazan L. Ali^a, Rawshan N. Othman^a, Hadil A. Hasan^a, Sara Azad^a, Naz A. Mahmood^a, Sivan S. Abdalrahman^a, Hezha O. Rasulⁱ, Nebojsa Bacanin^j, **S. Vimal^j**

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Abstract

One of the popular metaheuristic search algorithms is Harmony Search (HS). It has been verified that HS can find solutions to optimization problems due to its balanced exploratory and convergence behavior and its simple and flexible structure. This capability makes the algorithm preferable to be applied in several real-world applications in various fields, including healthcare systems, different engineering fields, and computer science. The popularity of HS urges us to provide a comprehensive survey of the literature on HS and its variants on health systems, analyze its strengths and weaknesses, and suggest future research directions. In this review paper, the current studies and uses of harmony search are studied in four main domains. (i) The variants of HS, including its modifications and hybridization. (ii) Summary of the previous review works. (iii) Applications of HS in healthcare systems. (iv) And finally, an operational framework is proposed for the applications of HS in healthcare systems. The main contribution of this review is intended to provide a thorough examination of HS in healthcare systems while also serving as a valuable resource for prospective scholars who want to investigate or implement this method.

Section snippets

Background and introduction

Before the year 2001, several search algorithms were already introduced and applied in different fields of applications. HS is one of the search algorithms that Zong Woo Geem has introduced. His vision was to develop a better-performing algorithm as an alternative to the other predefined algorithms. The intention of designing the algorithm is to have a generalized optimization technique for continuous, constrained, and discrete optimization in several types of optimization problems. Both the...

Current trends of HS



[International Journal of Ad Hoc and Ubiquitous Computing](#) > [2022 Vol.40 No.1/2/3](#)

Title: [Architecture and routing protocols for internet of vehicles: a review](#)

Authors: Farhana Ajaz; Mohd Naseem; Sparsh Sharma; Gaurav Dhiman; Mohammad Shabaz; S. Vimal

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Abstract: Modern vehicles should be able to commute a tremendous amount of data and information within their neighbourhood. To incorporate the requirements of modern vehicles, the conventional vehicular ad hoc network (VANETs) are emerging to the internet of vehicles (IoV). IoV keeps all the smart vehicles connected with the help of sensors, GPS, entertainment system, brakes and throttles. These devices send and store their data with the help of cloud. This paper intends to contribute to the review of IoV, its challenges, characteristics and application. A detail discussion on architectures and routing protocols along with its classification is also discussed. This paper ought to guide and motivate researchers working in the area of IoV to develop scalable and efficient routing protocols.

Keywords: internet of vehicles; IoV; internet of things; IoT; routing protocols; architecture; vehicular ad hoc networks; VANETs; mobile ad hoc networks; MANETs; cloud computing; fog computing.

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Review Article | Open Access

Volume 2022 | Article ID 7055910 | <https://doi.org/10.1155/2022/7055910>[Show citation](#)

Improved Fitness-Dependent Optimizer for Solving Economic Load Dispatch Problem

Barzan Hussein Tahir ¹, Tarik A. Rashid ¹, Hafiz Tayyab Rauf ², Nebojsa Bacanin ³, Amit Chhabra ⁴, S. Vimal ⁵ and **Zaher Mundher Yaseen**  ^{6,7,8}

[Show more](#)**Academic Editor:** Aboul Ella Hassanien**Published:** 11 Jul 2022

Abstract

Economic load dispatch depicts a fundamental role in the operation of power systems, as it decreases the environmental load, minimizes the operating cost, and preserves energy resources. The optimal solution to economic load dispatch problems and various constraints can be obtained by evolving several evolutionary and swarm-based algorithms. The major drawback to swarm-based algorithms is premature convergence towards an optimal solution. Fitness-dependent optimizer is a novel optimization algorithm stimulated by the decision-making and reproductive process of bee swarming. Fitness-dependent optimizer (FDO) examines the search spaces based on the searching approach of particle swarm optimization. To calculate the pace, the fitness function is utilized to generate weights that direct the search agents in the phases of exploitation and exploration. In this research, the authors have used a fitness-dependent optimizer to solve the economic load dispatch problem by reducing fuel cost, emission allocation, and transmission loss. Moreover, the authors have enhanced a novel variant of the fitness-dependent optimizer, which incorporates novel population initialization techniques and dynamically employed sine maps to select the weight factor for the fitness-dependent optimizer. The enhanced population initialization approach incorporates a quasi-random Sobol sequence to generate the initial



LAKE-BSG: Lightweight authenticated key exchange scheme for blockchain-enabled smart grids

Akhtar Badshah^a, Muhammad Waqas^{b, f}, Ghulam Abbas^{a, c}, Fazal Muhammad^d, Ziaul Haq Abbas^{a, c}, S. Vimal^g, Muhammad Bilal^h  

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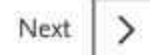
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Abstract

The Smart grid employs information and communication networks to collect data on energy production and consumption, and then utilize it to improve efficiency, reliability, economic benefit, power generation, and distribution sustainability. However, in SG networks the components communicate over insecure public channels, raising severe security concerns. Likewise, the security challenge is growing more difficult with the introduction of virtual solutions in SG networks. Thus, we devise a new scheme, named lightweight authenticated key exchange scheme for a blockchain-enabled SG environment (LAKE-BSG), which permits secure communication among smart meters (SMs) and service providers (SPs). In the proposed scheme, data is kept secured in a blockchain network. SPs are liable for verifying new blocks in the private blockchain through a consensus algorithm. We provide a detailed security analysis of LAKE-BSG through informal security analysis, formal security analysis using the real oracle model, and formal security verification using the Scyther tool. The results demonstrate that the devised scheme is resilient towards various security attacks in an SG environment. Furthermore, an exhaustive comparative analysis reveals that LAKE-BSG is efficient in terms of communication and computation overheads and provides additional security and functionality features.



Keywords

Smart grid; Blockchain; Key exchange scheme; Security; Authentication

Expert Systems / Early View / e13029

ORIGINAL ARTICLE

An AI powered system call analysis with bag of word approaches for the detection of intrusions and malware in Australian Defence Force Academy and virtual machine monitor malware attack data set

Appu Alfred Raja Melvin  Gnanaraj Jaspher W. Kathrine, Subbulakshmi Pasupathi, Vimal Shanmuganathan, Rajalingam Naganathan

First published: 18 May 2022

<https://doi.org/10.1111/exsy.13029>

Correction added on 27 May 2022 after first online publication: Author name has been corrected in this version.

Abstract

This study propose the use of AI enabled machine learning algorithms with the Bag-of-Word (BoW) methods for the detection of intrusions by analysing the system call patterns. Host based Intrusion Detection System can make use of system call patterns to differentiate between normal and anomalous program behaviours. First, the system call patterns are pre-processed with different approaches like BoW, BoW with Boolean value, BoW with Probability value and BoW with TF-IDF. Next machine learning algorithms are used to evaluate the performance of classifier models. We used J48 (C4.5), Random Forrest, RIPPER, KNN, SVM, and NaiveBayes ML algorithms. This process was carried out on ADFA-LD and on our proposed virtual machine monitor (VMM) malware attack data set for analysis. The proposed work is evaluated based on detection accuracy and false alarm rate metrics. Random Forrest algorithm performs better compared with other ML algorithms in terms of intrusion detection accuracy and false alarm rate on ADFA and VMM malware data set. The proposed data set provide better results compared with ADFA-LD analysed using ML algorithms. The classifier model trained with ADFA and VMM malware system call data sets may do predictive analytics in detecting security issues for Industry 4.0 systems.

CONFLICT OF INTEREST



Comparative analysis of FSO, OFC and diffused channel links in photonics using artificial intelligence based S-band, C-band and L-band of the attenuation metrics

G. Karpagarajesh¹ · R. Santhana Krishnan² · Y. Harold Robinson³ · S. Vimal⁴ · S. Thamizharasan⁵ · P. Subbulakshmi⁶ · M. Kaliappan⁴

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Abstract

Optical communication system is preferred over microwave and radio frequency communication system because of license free operation. Simulative analysis of 10gbps Optical Communication using different optical bands of L-Band S-Band, E-Band and C-Band under different attenuation condition has been performed in this paper. The Free Space Optics system has been analyzed with transmission distance of 1 km using Optical Wireless Communication channel and the effect of variation in attenuation value in the scenario of different weather conditions (clear, haze, rain and fog) were also examined. The Optical Fiber Communication is also examined under conditions of attenuations due to physical imperfections. The cross-layer Artificial Intelligence Framework is implemented to perform the prediction of different optical communication. The diffused channel is studied by



Bioinformatics and Medical Applications: Big Data Using Deep Learning Algorithms

A. Suresh (Editor)(/en-gb/search?pq=%7Crelevance%7Cauthor%3AA.+Suresh), S. Vimal (Editor)(/en-gb/search?pq=%7Crelevance%7Cauthor%3AS.+Vimal), Y. Harold Robinson (Editor)(/en-gb/search?pq=%7Crelevance%7Cauthor%3AY.+Harold+Robinson), Dhinesh Kumar Ramaswami (Editor)(/en-gb/search?pq=%7Crelevance%7Cauthor%3ADhinesh+Kumar+Ramaswami), R. Udendhran (Editor)(/en-gb/search?pq=%7Crelevance%7Cauthor%3AR.+Udendhran)

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[Manoj Diwakar](#), [Prabhishek Singh](#), [Achyut Shankar](#),
[Soumya Ranjan Nayak](#) , [Janmenjoy Nayak](#), [S. Vimal](#),
[Ravinder Singh](#) & [Dilip Sisodia](#)

[Network Modeling Analysis in Health Informatics and Bioinformatics](#) **11**, Article number: 15 (2022)

258 Accesses | **3** Citations | [Metrics](#)

Abstract

Smart healthcare is being adopted gradually as information technology advances. The enormous increase in demand for smart medical imaging has resulted in the fusion of a number of important imaging technologies. In smart imaging, many times single modality images are not sufficient to extract the major or minor information from medical images. Therefore in this paper, a new fusion algorithm is introduced for multi-modality medical images to extract maximum information and provide an efficient fused image. In proposed scheme, NSCT is used to get low- and high-frequency components of the medical images.

[Home](#) > [Soft Computing](#) > [Article](#)Focus | [Published: 20 April 2022](#)

A rough set theory and deep learning-based predictive system for gender recognition using audio speech

[Ghazaala Yasmin](#), [Asit Kumar Das](#) , [Janmenjoy Nayak](#), [S. Vimal](#) & [Soumi Dutta](#)[Soft Computing](#) (2022)183 Accesses | 3 Citations | [Metrics](#)

Abstract

Speech is one of the most delicate medium through which gender of the speakers can easily be identified. Though the related research has shown very good progress in machine learning, but recently, deep learning has imparted a very good research area to explore the deficiency of gender discrimination using traditional machine learning techniques. In deep learning techniques, the speech features are automatically generated by the reinforcement learning from the raw data which have more discriminating power than the human-generated features. But in some practical situations like gender recognition, it is observed that combination of both types of features sometimes

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Extreme learning machine and bayesian optimization-driven intelligent framework for IoMT cyber-attack detection

[Janmenjoy Nayak](#) , [Saroj K. Meher](#), [Alireza Souri](#),
[Bighnaraj Naik](#) & [S. Vimal](#)

The Journal of Supercomputing **78**, 14866–14891 (2022)

1825 Accesses | **10** Citations | [Metrics](#)

Abstract

The Internet of Medical Things (IoMT) is a bionetwork of allied medical devices, sensors, wearable biosensor devices, etc. It is gradually reforming the healthcare industry by leveraging its capabilities to improve personalized healthcare services by enabling seamless communication of medical data. IoMT facilitates prompt emergency responses and provides improved quality of medical services with minimum cost. With the advancement of modern technology, progressively ubiquitous medical devices raise critical security

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Autoencoder for Design of Mitigation Model for DDOS Attacks via M-DBNN

Ankit Agrawal ¹, Rajiv Singh ¹, Manju Khari ², S. Vimal ³ and Sangsoon Lim  ⁴[Show more](#)**Academic Editor:** Nima Jafari Navimipour**Published:** 07 Apr 2022

Abstract

Distributed Denial of Service (DDoS) attacks pose the greatest threat to the continued and efficient operation of the Internet. It can lead to website downtime, lost time and money, disconnection and hosting issues, and website vulnerability. Conventional machine learning methodologies are being harmed by reduced recognition rates and greater false-positive rates due to the emergence of new threats. As a result, high-performance machine learning classifiers with low false-positive rates and high prediction accuracy are required for the DDoS detection system. Here, a deep belief neural network is preferred, upgraded to the modified deep neural network (M-DBNN) to accurately detect DDoS attacks from the network. Enable the database to change a specific format and range, which helps the M-DBNN classifier easily predict the class. An advanced Chimp Optimization Algorithm (ChOA) is used to minimize the error to find the best weight of the M-DBNN classifier; this leads to accurate DDOS attack detection and predict the classes effectively. The proposed method is evaluated for CAIDA "DDoS Attack 2007" dataset. The accuracy of the proposed method is 0.87%, and the outcome is compared with those of other existing methods of deep neural network (DNN), support vector machine (SVM), artificial neural network (ANN), and neural network (NN). The proposed method demonstrates great detection accuracy with a low error.

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Abstract

The human body's major organ is the skin, and it protects human beings from the outside environment. Detecting skin disease at an earlier stage is a big challenge because of the similar appearance of skin disease. Although skilled dermatologists find it challenging to forecast skin lesions due to lack of contrast between adjoining tissues. Therefore, there is a need for an automated system that can detect skin lesions timely and precisely. Recently Deep Learning (DL) has attained outstanding success in the diagnosis of various diseases. Thus, in this paper, a transfer learning-based model has been proposed with help of pre-trained Xception model. The Xception model was modified by adding layers such as one pooling layer, two dense layers and one dropout layer. A new Fully Connected (FC) layer changed the original Fully Connected (FC) layer with seven skin disease classes. The proposed model has been evaluated on a HAM10000 dataset with large class imbalances. The data augmentation techniques were applied to overcome the unbalancing in the dataset. The new results showed that the model has attained an accuracy of 96.40% for classifying skin diseases. The proposed model is working best on Benign Keratosis and the values of precision, sensitivity and F1 score are 99%, 97% and 0.98 respectively. This method can provide patients and doctors with a good notion of whether or not medical assistance is required, thus, avoiding undue stress and false alarms.

Keywords: Deep learning ▪ skin disease classification ▪ HAM10000 dataset ▪ transfer learning ▪ Xception ▪ convolutional neural network ▪ image analysis

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Abstract:

The Internet of Medical Things (IoMT) has recently gained a lot of interest in the health care industry. IoMT enables real-time and omnipresent monitoring of a patient's health status, resulting in massive amounts of medical data being generated. The centralized massive data processing places enormous strain on the typical cloud computing, rendering it incapable of supporting a variety of real-time health care applications. Therefore, edge computing that moves application programs and data processing from central infrastructure to the edge nodes has attracted wide attention. However, adopting existing edge server (ES) deployment strategies for IoMT is not suitable due to the decentralized and high real-time service requirements of IoMT systems. In particular, traditional ES deployment strategies in IoMT system confront major load imbalance across ESs, latency issues, and energy consumption concerns. To address these challenges, a deployment strategy of ESs based on the state-action-reward-state-action (SARSA) learning, named ESL, is designed. Specifically, ESs are quantified by evaluating the silhouette coefficient (SC) and the sum of squared errors. Then, through fuzzy C-means (FCM) algorithm, the preliminary division of health monitoring units (HMUs) and the initial locations of ESs are obtained. Finally, SARSA learning is adopted to determine the deployment of ESs. Furthermore, extensive experiments and analyses confirm that ESL achieves the core objective of optimizing load balancing among ESs while also optimizing request-response latency and request processing energy consumption.

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Multi-class Skin Disease Classification Using Transfer Learning Model

Vatsala Anand, Sheifali Gupta, Deepika Koundal, Soumya Ranjan Nayak, Janmenjoy Nayak  and **S. Vimal**

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Abstract

The human body's major organ is the skin, and it protects human beings from the outside environment. Detecting skin disease at an earlier stage is a big challenge because of the similar appearance of skin disease. Although skilled dermatologists find it challenging to forecast skin lesions due to lack of contrast between adjoining tissues. Therefore, there is a need for an automated system that can detect skin lesions timely and precisely. Recently Deep Learning (DL) has attained outstanding success in the diagnosis of various diseases. Thus, in this paper, a transfer learning-based model has been proposed with help of pre-trained Xception model. The Xception model was modified by adding layers such as one pooling layer, two dense layers and one dropout layer. A new Fully Connected (FC) layer changed the original Fully Connected (FC) layer with seven skin disease classes. The proposed model has been evaluated on a HAM10000 dataset with large class imbalance. The data



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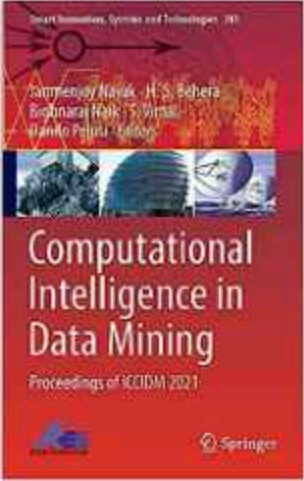
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